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DSTALGO
ANSWER SHEET
Term 1 AY 2026 - 2027

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Data Structures and Algorithms

- A **data structure** is a way of storing and organizing data so that it can be used efficiently. Different data structures are used depending on the type of data and what operations need to be performed on it. Common examples include arrays, linked lists, stacks, queues, trees, and graphs.

Data structures help programs manage large amounts of data efficiently. They make it easier to store, access, organize, and manipulate information. Choosing the appropriate data structure can help a program use less memory and perform tasks more efficiently.

- An **algorithm** is a set of step-by-step instructions used to solve a specific problem or achieve a particular goal. In computer science, algorithms operate on data structures to perform tasks such as searching, sorting, and manipulating data.

Algorithms provide a systematic way of solving problems. An efficient algorithm can make a program faster and help it use resources more effectively. Examples include algorithms for searching, sorting, finding routes, and processing data.

Linear Data Structure

A **linear data structure** is a type of data structure where elements are arranged in a sequential order, one after another. This organization makes it easier to access and process data in a specific sequence.



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- Linear data structures provide a simple way to store, access, and manage data.
- They are commonly used in different algorithms and computer applications.
- Examples include arrays, linked lists, stacks, and queues.
- In contrast, nonlinear data structures organize data in a hierarchical or interconnected way, such as trees and graphs.

Overall, linear data structures are important because they provide an organized and straightforward way to handle data and serve as a foundation for many algorithms and applications.

One-Dimension Array

A ***one-dimensional array*** is a linear data structure that stores elements of the same data type in contiguous memory locations. It provides a systematic way of organizing and accessing a collection of elements, where each element is identified by its index or position within the array. The index starts from 0 and goes up to the size of the array minus one.

One-dimensional arrays are useful and robust data formats that can be used for a variety of data storage and manipulation purposes. They are used in a variety of applications and are a necessary component of many C programs.

The Big O Notations

Big O notation is used to describe the time or space complexity of algorithms. Big-O is a way to express an upper bound of an algorithm's time or space complexity.



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- Describes the asymptotic behavior (order of growth of time or space in terms of input size) of a function, not its exact value.
- Can be used to compare the efficiency of different algorithms or data structures.
- It provides an** upper limit** on the time taken by an algorithm in terms of the size of the input. We mainly consider the worst case scenario of the algorithm to find its time complexity in terms of Big O

Basic Searching Algorithms

Searching algorithms are methods used in computer science to find a specific item within a collection of data, such as an array. Two common searching algorithms are Linear Search and Binary Search.

- A **Linear Search** is commonly used for an unsorted array. It checks each element one by one until the target item is found or the end of the array is reached. Its time complexity is $O(n)$.

It is a simple and straightforward method that is useful when the data is small or not sorted. It is easy to implement but can take longer when searching through a large dataset.

- A **Binary Search** is used for a sorted array. It checks the middle element first and then searches either the left or right half depending on the comparison. By repeatedly dividing the search area in half, it has a time complexity of $O(\log n)$.

It is an efficient searching method because it eliminates half of the remaining elements after each comparison. This makes it faster than Linear Search when working with large, sorted datasets.



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